

Akotet Yeshaw Tesema

akiyeshaw@gmail.com | +82-10-9703-0616 | akotet08.github.io | [akotet-yeshaw](https://akotet-yeshaw.github.io)

Education

KAIST – B.S. in Computer Science (Minor in AI) Sep 2021 – Aug 2025
Selected coursework: Statistical Learning Theory, Operations Research: Optimization, Algorithms, Operating Systems, Deep Learning, ML for Computer Vision, Data Science, Wearable UIs

KAIST – M.S., School of Electrical Engineering Sep 2025 – Present
Research Area: Recommender Systems, Graph Neural Networks, Machine Learning

Publications

-
- **Akotet Yeshaw Tesema**, Biniyam Aschalew Tolera, Cheonwoo Lee, Jaemin Yoo. “Beyond Node Degree: Learning-Driven Tailness for Graph Collaborative Filtering.” *Under review at ACM SIGKDD (KDD) 2027*.

Research Experience

Visiting Researcher | Seoul National University Mar 2026 – Present
Seoul, Korea

- Developing a **unified foundation model for generative recommendation**, targeting in-context item-token generation across datasets.
- Designing the end-to-end model architecture and training recipe under the supervision of Prof. Jaemin Yoo.

Research Fellow | Equitech Futures 2026 – Present
Remote

- Collaborating with faculty and associate fellows on research into **sycophancy in Large Language Models (LLMs)** – models over-agreeing with users at the expense of accuracy.
- Designing evaluations and mitigation strategies to measure and reduce sycophantic behavior in LLM responses.

Applied Data Institute Scholar | Equitech Futures Oct – Dec 2025
Remote

- Selected as a Top 20 scholar from 200+ applicants across 80+ countries in a competitive global program.
- Collaborated in an international team to design a data-driven water utility dashboard for selected African countries.
- Conducted parallel research on **conversational echo chambers in Large Language Models (LLMs)**.

Undergraduate Research Intern | Data AI Lab, KAIST Sep 2023 – Aug 2025
Daejeon, Korea

- Implemented and tested GNN baselines across multiple datasets; contributed experiments and ablations.
- Led a project to mitigate popularity bias in graph-based recommender systems.

AI Researcher | Enhans (startup) Jan – Mar 2025
Seoul, Korea

- Built an automatic prompt-optimization pipeline for LLMs, boosting answer coherence by **10%**.
- Designed a synthetic reasoning-chain generator for iterative prompt refinement.

Research Intern | Networking & Mobile Systems Lab, KAIST Jun – Sep 2024
Daejeon, Korea

- Formulated and evaluated baselines for multivariate time-series imputation with missing modalities.
- Built a dynamic imputation framework for generating missing data points.

Research Intern | SMILE Lab, KAIST Jun – Sep 2023
Daejeon, Korea

- Optimized Hawkeye localization to **sub-cm accuracy** over 100 m using Fourier-domain filtering.
- Distinguished mobile vs. static tags, enabling dynamic power scaling.

Teaching & Mentoring

Teaching Assistant | AddisCoder

Summer 2026

Addis Ababa, Ethiopia

- Mentored Ethiopian high-school students in Python programming and algorithm design during an intensive tuition-free summer program.
- Led daily lab sessions, guiding students through problem-solving and hands-on coding exercises.

Selected Projects

Ad Recommendation – Graph-Based Click & Ranking Model | Personal Project

2026

- Built a GCN ad recommender on a heterogeneous user-search-ad graph (UniMP-inspired) with a LightGCN baseline.
- Jointly optimized click prediction (F1) and ad ranking (NDCG@3) with negative sampling; tracked via Weights & Biases.

Multi-Behavior Recommendation – NGCF | Personal Project (GitHub)

2025

- Modeled hierarchical view/save/purchase signals over a 21K-user benchmark with an NGCF backbone.
- Jointly handled interaction-type classification (Macro-F1) and top-K ranking; benchmarked against LightGCN and MuLe.

Water Utility Performance Platform | Applied Data Institute (GitHub)

2025

- Built Streamlit dashboards with role-based access and AI-generated insights for water & sanitation KPIs.
- Enabled national-, city-, and zone-level monitoring from a shared data pipeline.

PromptOpt – LLM Prompt-Optimization Service | Personal Project

2025

- Built a Dockerized critique-and-refine prompt optimizer (PromptWizard) over OpenAI and Anthropic backends.
- Deployed on AWS with SQS-driven asynchronous job processing.

Harvest ROS – Autonomous Fruit Harvester | Personal Project (GitHub)

2023

- Built an end-to-end robot on TurtleBot3 for strawberry picking.
- Fine-tuned YOLOv5 for six-level ripeness classification; fused RGB and ultrasonic depth.

ModiCharachorder – Custom Input Device | Personal Project (GitHub)

2022

- 12-button + IMU device with TensorFlow GRU autocomplete.

IP & Software Assets

Community Energy-Mix Optimization System | Software Registration

Jun 2024

Korea

- Web app recommending regional renewable portfolios; led a cross-functional team and built the optimization engine.

Building Remodeling Information & Case-base System | Software Registration

Jan 2024

Korea

- Designed the data model and cost-benefit algorithms for a sustainable retrofit database.

Honors & Leadership

-
- Aspire Leaders Program Alumnus (2025)
 - Undergraduate Research Participation Excellence Award (2024) – AI pest-attractant surveillance.
 - Top 5 / 180 teams – Korea Customs Data-Science Challenge (2023; fake-import detection).
 - Web Developer – KAIST International Students Association (2021–2023).

Technical Skills

Languages: Python | C | Java | JavaScript | Bash

ML Frameworks: PyTorch | TensorFlow

Tools: ROS | Django | Git | Linux